JERIN JACK JOSEPH RAJ

Software Engineer – Java, Spring Boot & Microservices

United States | Open to Relocation | jerinjack28@gmail.com

LinkedIn | GitHub | Portfolio

PROFESSIONAL SUMMARY

Software Engineer with 4+ years of professional Java development experience building scalable Spring Boot microservices and RESTful APIs. Skilled in Java 8/11/17, SOLID design principles, and relational and NoSQL database design across Oracle, MySQL, PostgreSQL, and MongoDB. Experienced building and consuming REST APIs, working with Kafka and JMS for distributed messaging, and implementing Spring Security, OAuth2, and JWT for API-level security. Uses AI coding assistants as part of the daily development workflow to improve code quality and productivity. Proficient with Git, CI/CD pipelines (Jenkins, GitHub Actions), Docker, and working knowledge of Kubernetes and AWS/Azure cloud platforms. Strong communicator with experience in code reviews, technical documentation, and cross-functional collaboration in Agile/Scrum environments, with a track record of debugging production issues and improving system performance.

TECHNICAL SKILLS

Languages & Core: Java 8/11/17, Core Java, Object-Oriented Programming (OOP), SOLID Principles, SQL

APIs & Microservices: RESTful APIs, gRPC (working knowledge), Microservices Architecture, API Design, HTTP Protocols

Frameworks & Security: Spring Boot, Spring Framework, Spring MVC, Spring Security, Spring Data, Hibernate/JPA, OAuth2, JWT

Databases: Oracle (PL/SQL, query tuning), MySQL, PostgreSQL, MongoDB, schema/table design, indexing, SQL optimization

Messaging: Apache Kafka, JMS, distributed messaging, event-driven architecture

Cloud & DevOps: AWS SDK, Azure SDK, Docker, Kubernetes (working knowledge), Git, Jenkins, GitHub Actions, CI/CD Pipelines

Testing & Code Quality: JUnit, Mockito, Postman, Code Reviews, Debugging, Performance Tuning, Root Cause Analysis, SonarQube

AI-Assisted Development: AI coding assistants used in daily development workflow for code generation, test scaffolding, and code quality

Frontend (Full-Stack Exposure): Angular 8, HTML5, CSS3, TypeScript

Professional Skills: Communication, Technical Writing, Customer Service, Cross-Functional Collaboration

Methodologies: Agile/Scrum, Sprint Planning, Jira, Code Reviews, Design Discussions

PROFESSIONAL EXPERIENCE

AVPS Technologies Inc – *Software Engineer* **Jul 2026 – Present**

Alpharetta, GA

- Delivered 12+ production REST endpoints for the company's content and analytics platform by designing and developing Java 8/17 Spring Boot microservices with a clean controller-service-repository architecture and reusable DTOs.
- Reduced response times on high-traffic endpoints by approximately 25% by resolving a Hibernate mapping issue, adding missing database indexes, and optimizing SQL queries against Oracle and PostgreSQL.
- Reduced service coupling and improved reliability by implementing Kafka and JMS-based asynchronous processing for audit events and workflow notifications.
- Maintained zero production defects over the past year by conducting code reviews and mentoring two junior developers on engineering standards.
- Reduced manual deployment effort by managing CI/CD pipelines in Jenkins and GitHub Actions, containerizing services with Docker, and deploying to AWS ECS using AWS and Azure SDKs.
- Improved day-to-day development velocity by applying AI coding assistants to boilerplate code generation, test scaffolding, and documentation.
- Supported end-to-end application workflows by integrating backend services with the Angular front end and downstream systems.

Chicago Education Advocacy Cooperative – *Software Engineer* **Aug 2025 – Jul 2026**

Chicago, IL

- Built out the backend for the organization's content and internal operations tools by developing a set of Java 17 Spring Boot services with reusable DTO contracts.
- Cut a recurring reporting query from several seconds to under a second by adding the right database indexes and rewriting an inefficient join.
- Reduced dependency on synchronous service calls by implementing Kafka-based asynchronous processing for audit events and workflow updates.
- Improved release quality by writing unit and integration tests using JUnit, Mockito, and Postman, and raising SonarQube code coverage ahead of releases.
- Resolved Tier 3 production issues faster by using Log4j output and API traces for diagnosis and root cause analysis.

- Maintained consistent, reliable deployments by managing CI/CD pipelines with GitHub Actions, Jenkins, and Docker on AWS ECS/Kubernetes-compatible infrastructure.
- Supported cross-team API integration by producing OpenAPI-style documentation alongside PostgreSQL schema and query development.
- Contributed to consistent sprint delivery by participating in sprint planning, story estimation, and code review with QA, DevOps, and product teams.

Gulf International Bank (TCS) – *Backend Developer* **Oct 2021 – Jul 2023**

- Delivered core backend services for balance transfers, international payments, account lookups, and reconciliation by developing Java and Spring Boot applications for a regulated banking client.
- Cut peak-time payment API response times nearly in half by tuning Oracle/PL-SQL queries, adding indexes, managing transactions, and implementing Redis caching.
- Enabled secure REST and SOAP service integration by building request validation, response mapping, and error-code handling with Spring MVC and Hibernate/JPA.
- Extended payment processing reliability by integrating internal systems with partner APIs and core banking platforms through WSO2 API Manager, JMS, and Kafka event flows.
- Supported nightly reconciliation and business reporting by building Apache Spark batch jobs and ETL pipelines feeding Tableau dashboards.
- Reduced release risk by conducting code reviews, mentoring junior developers, and maintaining Jenkins CI/CD pipelines, with sprint tracking in Jira.

Iota Automation Private Limited – *Java Developer* **Apr 2020 – Jul 2021**

- Delivered REST APIs and backend services for internal enterprise applications by developing Java and Spring Boot solutions on Oracle and MySQL.
- Improved code maintainability by applying SOLID principles and standard design patterns, including DTOs, the repository pattern, and centralized exception handling.
- Sped up high-read reporting operations by optimizing queries and adding targeted database indexes.
- Reduced regression risk by writing unit and service-level tests using JUnit and Mockito and validating APIs with Postman.
- Supported production deployments by managing Linux/Unix environments, Tomcat configuration, and shell scripts for build validation.

RELEVANT PROJECTS

Vehicle Reservation Microservice & Angular Front-End

- Delivered an end-to-end vehicle reservation platform by developing a Java/Spring Boot backend and an Angular 8 front end for search, booking, and an administrative dashboard.
- Enabled real-time booking updates by integrating GIS-based location mapping and Azure IoT Hub/Event Hub for vehicle telemetry.
- Secured the platform's API layer by implementing JWT authentication, role-based access control, and JUnit/Mockito test coverage, deployed via Docker/Kubernetes through Jenkins CI/CD.

AI-Powered Document Processing Platform

- Automated processing of invoices, IDs, and contracts by building a Spring Boot service that uses AI-based text extraction and NLP classification models.
- Improved processing throughput by implementing Kafka-based asynchronous job queuing, with results stored across MongoDB and Oracle by document type.
- Enabled reliable integration for downstream consumers by exposing REST APIs for upload, status tracking, and result retrieval, backed by unit/integration tests and SonarQube quality gates.

Real-Time Analytics & ETL Reporting Pipeline

- Enabled real-time business reporting by building a Kafka and Spark streaming pipeline into Oracle-based reporting tables.
- Delivered stakeholder-facing operational dashboards by building ETL jobs to clean, transform, and load multi-source data, published via Tableau.

EDUCATION

Master of Information Technology and Management – Illinois Institute of Technology, Chicago, IL | GPA: 3.9